Abd Elkarim Amra

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Education

Macalester College St. Paul, MN

B.A. in Computer Science and Economics — GPA: 3.93/4.0 *Sept. 2023 – May 2027*

- **Honors & Awards:** Kofi Annan & UWC Davis Scholar, Flemming First Generation Scholar, Dean's List
- **Relevant Coursework:** Core Concepts of Computer Science, Object-Oriented Programming, Data Structures and Algorithms, Functional Problem Solving, Discrete Mathematics, Calculus I

Projects

Full-Stack Web Development (Course Projects) June 2025 - Present

- Developed responsive UIs with HTML, CSS (Flexbox, Tailwind), and React (Context API, Router, forms).
- Currently deploying a task manager web application with authentication and full CRUD functionality. [GitHub Repo]
- Implemented RESTful APIs with Node.js & Express and integrated MongoDB/Mongoose for schema design, validation, and data management.

Cafeteria Menu GUI Application (Java) April 2025 - May 2025

- Collaborated with a team to build a Java GUI for Macalester students to browse, search, and rate cafeteria menus.
- Used a Trie for fast autocomplete search and Jsoup to scrape daily specials from the cafe website. [GitHub Repo]
- Designed an interactive interface with Swing, featuring dietary filters and voting buttons.

Gesture Recognition Algorithm Implementation Using Java February 2025

- Extended starter code to implement an interactive drawing interface with mouse handlers for gesture capture.
- Developed the core 1 dollar gesture recognizer methods, which resulted in accurate recognition. [GitHub Repo]
- Integrated template management and UI feedback to display recognized gestures and scores for better interaction.

Productivity Application (Java) April 2024

- Collaborated with a team to develop a multifunctional timer and to-do list application in Java.
- Implemented features like automatic task completion and reset functionality. [GitHub Repo]
- Designed and built an intuitive user interface using Java Swing for seamless interaction.

Breakout Game (Java) March 2024

- Developed a Java Breakout game with paddle control, ball physics, and collision detection.
- Applied object-oriented design, modularizing the game for scalability. [GitHub Repo]
- Improved gameplay with random ball angles, smooth animations, and dynamic bricks.

Leadership and Professional Development

Campus Center Manager (CCM), Macalester College August 2023-Present

- Supervise the daily operations of the campus center and provide event support.
- Staff the information desk, and assist with facility management.

Volunteer, Math Tutor at Shahd Organization, Lebanon July 2024

- Tutored refugee children at Wavel Camp in mathematics and led programming workshops, introducing students to Python using the Turtles module.
- Developed communication, leadership, and problem-solving skills while supporting children with their academic needs.

Skills

- **Frontend:** HTML5, CSS3 (Flexbox, Responsive Design, Tailwind), JavaScript (ES6+), React (Context API, Router, Forms).
- **Backend:** Node.js, Express, REST API design, Authentication/Authorization.
- **Databases:** MongoDB, Mongoose.
- **Tools:** Git/GitHub, Linux, Deployment (CI/CD basics).
- **Languages:** Python, Java, JavaScript, Racket, HTML, CSS.